$\Omega \to$ all possible outcome

σ-algebra

1) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

2) $A_1, A_2 \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ & $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$

Defenation

$P: \mathcal{F} \to R$

i) $P(A) \geq 0$ ii) $P(\Omega) = 1$ iii) If A_i are mutually disjoint $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

Properties

1) $P(A^c) = 1 - P(A)$ 2) $P(\phi) = 0$ 3) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

4) $P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_3 \cap A_1) + P(A_1 \cap A_2 \cap A_3)$

Disjoint cover

i) $B_i \cap B_j = \phi$

ii) $\bigcup_{i=1}^{n} B_i = \Omega$

then $\{B_1 \dots B_n\}$ disjoint cover of Ω

Conditional probability

$$P(A_1 | B) = \frac{P(A_1 \cap B)}{P(B)}$$

$$P(A) = \sum_{i=1}^{N} P(B_i) P(A|B_i)$$ $\to$ If $\{B_1, B_2 \dots B_n\}$ disjoint cover of Ω.

Independence of events

$$P(A \cap B) = P(A) P(B)$$

Random Variable

$\hookrightarrow$ function from $\Omega \to R$

RV is discrete RV if $\to$ R_x is countably finite or countably infinite

pmf $\to$ Real valued fn $f: R \to R$ $f_X(x) = \text{Prob}(X = x)$

Properties of pmf $\to$ Defining

i) $f(x) \geq 0$

ii) $\{x \mid f(x) > 0\}$ is a countable set

$R_x = \{x_1, x_2 \dots\}$

iii) $\sum_{x_i \in R_x} f_X(x_i) = 1$

CDF

RV → pmf → pdf

$F(t) = P(X \le t)$

Properties of CDF

i) $0 \le F(t) \le 1$

ii) $\lim_{t \to \infty^+} F(t) = 1$ and $\lim_{t \to \infty^-} F(t) = 0$

Imp
iii) F is non decreasing function

iv) F is right continous function

→ CDF is discontinous only at $t \in R_X$ when X is ~~is~~ discrete RV

→ From CDF to pmf (In case of discrete RV)

1) check the pts in R where CDF has jumps
2) These are pts in range of X
3) pmf at these pts is the jump size

Lebesgue Decomposition theorem

$F(x) = G(x) + H(x)$

$F(x) \rightarrow$ CDF

$G(x) \rightarrow$ continous

$H(x) \rightarrow$ step function

If $G(x) = 0$ → discrete r.v.

If $H(x) = 0$ → continous r.v.

If neither is zero → mixed r.v.

→ In case of discrete rvs

CDF → pmf → by taking differences $F(t_i) - F(t_j)$

Continous rv.

pdf ← $f(x) = \frac{d}{dx} F(x)$

$P(a \le x \le b) = \int_a^b f_x(x)\,dx$

$$\boxed{\int_{-\infty}^{\infty} f_X(x)\,dx = 1}$$

Expectation/Mean

$E(X) = \sum_{x \in R_X} x f_X(x)$ if x is discrete

$= \int_{x \in R_X} x f_X(x)\,dx$ if x is continous

In general.

$E(h(x)) = \Sigma h(x) f_x(x)$ X is discrete

$= \int h(x) f_x(x) dx$ X is continous

$E(X^r) \rightarrow r^{th}$ raw moment $E(X-a)^m \rightarrow m^{th}$ central moment

$E(X-\mu)^r = \Sigma (x-\mu)^r f_x(x)$ discrete

$= \int (x-\mu)^r f_x(x) dx$ continous

when $r=2 \Rightarrow E(X-\mu)^2 =$ Variance of $X = Var(X) = \sigma^2$ (standard deviation)^2

$$\boxed{Var(X) = E(X^2) - [E(X)]^2}$$

$E(e^{tx}) = M_X(t) \rightarrow$ moment generating function $= \sum_{x_j \in R_X} e^{tx_j} f(x_j)$ discrete

$= \int_{x \in R_X} e^{tx} f(x) dx$ if X is continous

$\frac{d^r}{dt^r} M_X(t) \Big|_{t=0} = E(X^r)$ $\boxed{M_{aX+b}(t) = e^{bt} M_X(at)}$

$\boxed{\text{Median} \quad F_X(m) = \frac{1}{2}}$

Percentiles

$F_X(p_i) = \frac{i}{100} \rightarrow$ a pt p_i such that

2nd quartile = 50^{th} percentile

Properties of expectation

1) $P(X=\alpha) = 1 \quad E(X) = 1$
2) $E(cX) = cE(X)$
3) $E(X+Y) = E(X) + E(Y)$
4) If $P(X \geq Y) = 1$ then $E(X) \geq E(Y)$
5) $|E(X)| \leq E(|X|)$

Chebyshev's Inequality

$$P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2} \quad t > 0$$

Quartiles

$q_1 \rightarrow 1^{st}$ quartile $F_X(q_1) = \frac{1}{4}$

$q_2 \rightarrow 2^{nd}$ quartile or median $F_X(m) = \frac{1}{2}$

Mode $\rightarrow$ X be rv

$M = \arg\max f_x(x)$

Imp Memoryless Property

$P(X > x_0 + y_0 \mid X > x_0) = P(X > y_0)$ $\rightarrow$ followed by exponential and discrete geometric

$\hookrightarrow$ used in electrical devices

$f_X(x) \to$ pdf

$Y = \phi(X)$

$g(y) = f(\phi^{-1}(y)) \left| \frac{d}{dy}\phi^{-1}(y) \right|$ $\to$ Transformation

Symmetric densities

$f_X(x) = f_X(-x)$

$\boxed{F_X(-x) = 1 - F_X(x)}$ $\to$ for symmetric $\boxed{F_X(0) = \frac{1}{2}}$

Bernoulli

$f_X(x) = p \quad x=1$

$1-p \quad x=0$

$0 \quad$ o.w

Binomial r.v $\rightarrow M_X(t) = ((1-p)+pe^t)^n$

$\rightarrow E(X) = np$

↳ n independent bernoulli trials

$X \rightarrow$ no. of successes in n trials

$f_X = 0 \quad$ o.w

$\binom{n}{x} (p)^x (1-p)^{n-x} \quad x \in R_X = [0,1,2 \ldots n]$

Geometric $\rightarrow E(X) = \frac{1-p}{p}$

↳ independent bernoulli until first success

$f_X(x) = p(1-p)^x$

$X \rightarrow$ no. of failures before first success.

MGF $\rightarrow \frac{p}{1-(1-p)e^t}$

CDF $= 1-(1-p)^{[t]+1}$

Negative binomial

↳ perform independent bernoulli trials until first α no. of success

$f(x) = \binom{\alpha+x-1}{x} p^\alpha (1-p)^x$

Uniform continuous $X \sim U[a,b]$

$f_X(x) = \frac{1}{b-a} \quad a \le b \le b$

$F_X(x) = \frac{x-a}{b-a} \quad a < x < b$

$1 \quad x \ge b$

$E(X) = \frac{1}{2}(a+b)$

Cauchy RV

$f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2} \quad -\infty < x < \infty$

$E(X) \rightarrow$ does not exist

Exponential

$F_X(x) = \lambda e^{-\lambda x} \quad x>0$

$=0 \quad$ ow

$E(X^r) = \frac{r!}{\lambda^r} \quad m = \frac{-1}{\lambda} \log \frac{1}{2}$

Gamma

$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ for $x>0$

$E(X) = \frac{\alpha}{\lambda} \quad Var = \frac{\alpha}{\lambda^2}$

$E(X^r) = \frac{\alpha(\alpha+1)\cdots(\alpha+r-1)}{\lambda^r}$

Gam

Poisson

$f_X = \frac{e^{-\lambda}\lambda^x}{x!} \quad x=0,1,2 \ldots$

$0 \quad$ ow

$M_X(t) = e^{-\lambda} e^{\lambda e^t}$

Hypergeometric

$$p(x) = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}$$

$$E(x) = n\left[\frac{D}{N}\right]$$

$$Var(x) = n\left[\frac{D}{n}\right]\left[1-\frac{D}{n}\right]\left[\frac{N-n}{N-1}\right]$$

Poisson → counting no. of occurrences of an event in given unit of time

X → no. of events in fixed unit of time

Weibull distribu

Log Normal density

Mean square error

Hypergeometric

Poisson process

① Uniform ($U[a,b]$)

Discrete → probability $= \frac{1}{\text{no. of elements}}$

Continous

PDF $= \begin{cases} 1/b-a & x \in [a,b] \\ 0 & \text{ow} \end{cases}$

CDF $= \begin{cases} 0 & \\ \frac{x-a}{b-a} & x \in [a,b] \\ 1 & x > b \end{cases}$

$E(X) = \frac{a+b}{2}$ Var $(x) = \frac{(b-a)^2}{12}$

MGF $= \frac{e^{tb} - e^{ta}}{t(b-a)}$

Negative Binomial

$\binom{\alpha+x-1}{x} p^{\alpha}(1-p)^{x}$

↳ X → no. of failures before first α successes.

Mean : $\frac{\alpha p}{1-p}$

② Bernoulli (Discrete) → put $n=1$ for bernoulli

PMF $= p^{x}(1-p)^{1-x}$

CDF $= (1-p)^{1-x}$

$E(x) = p$

Var $(X) = p(1-p)$

$M_x(t) = 1-p+pe^{t}$

③ Binomial [$X \sim Bin(n,p)$] → n independent bernoulli trials are performed

PMF ${}^{n}C_{x}\, p^{x}(1-p)^{n-x}$

$E(X) = np$

$var(x) = np(1-p)$

MGF $= (1-p+pe^{t})^{n}$

X → no. of success in n trials

④ Geometric [$x \sim geo(p)$] → Memoryless

PMF $p(1-p)^{x}$

$E(X) = 1/p$

$Var(x) = 1-p/p^2$

mgf $= \frac{pe^{t}}{1-(1-p)e^{t}}$

→ X = No. of failures before first success

⑤ Poisson $[X \sim Po(\lambda)]$

PMF $= \frac{\lambda^x e^{-\lambda}}{x!}$ → 0 to ∞

$X \to$ no. of ~~other~~ events occuring in a fixed period of time

$E(X) = \lambda$ → Discrete

$Var(X) = \lambda$

MGF $= e^{\lambda(e^t - 1)}$

⑥ Standard Normal distribution $[X \sim N[\mu, \sigma^2]]$

PDF : $\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

$E(X) = \mu$

$Var(X) = \sigma^2$

$M_x(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$

⑦ Exponential → Memoryless property $X \sim \exp(\lambda)$

PDF $= \cancel{\frac{1}{\lambda} e^{-\lambda x}} = \lambda e^{-\lambda x}$ $[0, \infty)$

CDF $= \cancel{e^{-\lambda x}} = 1 - e^{-\lambda x}$

$E(X) = \cancel{\lambda}$ $1/\lambda$

$Var(X) = \cancel{\lambda^2}$ $1/\lambda^2$

$M_x(t) = \frac{\lambda}{\lambda - t}$

⑧ Gamma $[X \sim \text{gamma}(\alpha, \lambda)]$ → Continuous

PDF: $\left\{ \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \quad x > 0 \right.$

$E(X) = \frac{\alpha}{\lambda}$ ~~Var~~ $\cancel{\frac{\alpha}{\lambda^2}}$ $Var(X) = \frac{\alpha}{\lambda^2}$ $M_x(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha$

(5) Poisson $[X \sim Po(\lambda)]$

$x \rightarrow$ no. of events occurring

(9) Beta $X \sim Beta[\alpha, B]$

PDF : $\dfrac{\Gamma(\alpha+B)}{\Gamma(\alpha)\Gamma(B)} x^{\alpha-1}(1-x)^{B-1} \Rightarrow \dfrac{x^{\alpha-1}(1-x)^{B-1}}{B(\alpha, B)}$

$E(X) = \dfrac{\alpha}{\alpha+B}$ $\quad Var(X) = \dfrac{\alpha B}{(\alpha+B)^2(\alpha+B+1)}$

$B(\alpha, B) \rightarrow \dfrac{\Gamma(\alpha)\Gamma(B)}{\Gamma(\alpha+B)}$

Properties

$\Gamma(n) = (n-1)!$

$\Gamma(n+1) = n\Gamma(n)$

$\Gamma(1/2) = \sqrt{\pi}$

$\int_0^\infty x^{n-1} e^{-\lambda x} = \dfrac{\Gamma(n)}{\lambda^n}$ $\quad \Gamma(z) = \int_0^\infty x^{z-1} e^{-x} \cdot dx$

(5) Poisson $[X \sim Po(\lambda)]$

$X \Rightarrow$ no. of ~~atthe~~ events occuring

(9) Beta $X \sim Beta[\alpha, B]$

$$PDF : \frac{\Gamma(\alpha+B)}{\Gamma(\alpha)\Gamma(B)} x^{\alpha-1}(1-x)^{B-1} \Rightarrow \frac{x^{\alpha-1}(1-x)^{B-1}}{B(\alpha, B)}$$

$$E(X) = \frac{\alpha}{\alpha+B} \qquad Var(X) = \frac{\alpha B}{(\alpha+B)^2(\alpha+B+1)}$$

$$B(\alpha, B) \hookrightarrow \frac{\Gamma(\alpha)\,\Gamma(B)}{\Gamma(\alpha+B)}$$

Properties

$$\Gamma(n) = (n-1)!$$

$$\Gamma(n+1) = n\,\Gamma(n)$$

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\int_0^\infty x^{n-1} e^{-\lambda x} = \frac{\Gamma(n)}{\lambda^n}$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} \cdot dx$$